INCH-POUND

MIL-PRF-29595/2(AS)
w/AMENDMENT 2
10 April 2023
SUPERSEDING
MIL-PRF-29595/2(AS)
w/AMENDMENT 1
24 October 2016

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 26-VOLT, 50-AMPERE-HOUR, LITHIUM RECHARGEABLE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-29595.

REQUIREMENTS:

1. Part or identifying number. The part or identification number (PIN) of the battery covered by this specification sheet shall be M29595/2-1.

2. Dimensions and weight. The dimensions and configuration of the battery shall be as shown on figure 1 contained in this specification sheet. The weight of the battery shall be not greater than 53 pounds (24.1 kg).

3. Capacity. The initial capacity shall be not less than 55 ampere-hours (Ah). The 1C-rate capacity (see 6.14.2 and 6.14.9) shall be not less than 50 ampere-hours at the end of life. The battery shall provide not less than 50 Ah capacity during the final discharge of the life cycle test based on a 50 ampere, constant current discharge for 1 hour, at 24 °C (75 °F) ambient temperature, to a cutoff potential of 20.0 volts (50 Ah/1.0 hr/24 °C/20.0 V at end of life).

4. The container handles shall be part number 955-SS-SS-2-RG (manufactured by Nielsen Hardware, Zero Corporation, Hartford, CT 06141) or equivalent.

5. The J2 connector receptacle shall be in accordance with MIL-DTL-26482, Series II and shall mate with a connector plug conforming to MS3476L10-6S or equivalent. It shall be located as shown on figure 1. The shell and contact pins shall be constructed of a nickel-alloy material such as Inconel. The connector receptacle need not be hermetically sealed.

a. Pin A: center tap voltage
b. Pin B: temperature sensor negative

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c. Pin C: temperature sensor positive
d. Pin D: thermal switch
e. Pin E: thermal switch
f. Pin F: spare

6. Qualification, conformance and safety inspection. Table I of this specification sheet lists the required qualification and conformance inspections. Table II lists the required S9310 AQ SAF 010 tests to be performed before the manufacturer can receive safety certification approval and qualified products database (QPD) listing.

7. MIL-PRF-29595 variance. The battery shall comply with MIL-PRF-29595 except as follows:

7.1 The following paragraphs are not applicable:

3.8.2 Thermal protection.
3.9.4 Dielectric strength and insulation resistance.
3.9.5 Vent tubes.
3.9.11 Strength of receptacle.
3.9.2 Transient drop test.
4.5.5 Dielectric strength and insulation resistance.
4.5.6 Strength of vent tubes test.
4.5.10 Strength of receptacle test.
4.5.16 Short circuit test.
4.5.17 Thermal protection.
4.5.24 Wormgear retaining pin hardness.
4.5.27 Transient drop test.

7.2 Modify the following subparagraphs:

3.5 Design. Delete the text and insert: “The design shall conform to the requirements specified by MIL-PRF-29595, and MIL-PRF-29595/2 when examined in accordance with the visual and mechanical tests of 4.5.2 of MIL-PRF-29595. The design shall also conform to the requirements specified in S9310-SAF-AQ-010 (see 6.17 and 6.18 of MIL-PRF-29595).”

3.5.3 Battery container and cover. Delete the text in the second sentence and insert: “The dimensions and locations of the receptacle, vent tubes, and battery identification shall be as shown on figure 1 herein.”

3.5.4 Venting. Delete the fourth sentence.

3.5.5 Receptacles. Following the last sentence, add: “The J1 receptacle shall conform to MIL-PRF-18148/3 and be located as shown on figure 1 herein.”

3.5.9 Heaters. Delete the second sentence and insert: “Heater power shall not be provided during any discharge performed in section 4, except for 4.5.21c.” Delete 3.5.9.a and insert “Heaters shall be powered by a power source internal to the battery container.”

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3.7.1 Color. Delete the text and insert: "The color under natural light of all visible parts of each battery, except the mounting points, other external hardware, identification marking, and instructions, shall conform to the color 17875 (white), in accordance with SAE AMS-STD-595."

3.7.3 Identification marking. Delete the text in the third sentence and insert the following: "The marking shall contain the required titles and information as shown on figure 1 of MIL-PRF-29595 and shall be placed in accordance with label #1 as shown on figure 1 herein."

3.7.4 Battery caution marking. Delete the text in the first sentence and insert the following: "The battery caution marking shall contain the required titles and information as shown in figure 2 of MIL-PRF-29595 and shall be placed in accordance with label #2 shown on figure 1 herein."

3.9.6 State of charge indicator. Delete the text and insert: "A state of charge device with indicator having no less than four segments and communications bus shall be incorporated and be as shown on figure 1 herein."

3.9.8 Open circuit voltage. Delete the text and insert: "The open circuit voltage shall not be greater than 29 volts when tested in accordance with 4.5.4 of MIL-PRF-29595."

3.9.9 Capacity and electrical performance. Delete the entire text and insert: "The initial capacity shall be not less than 55 ampere-hours. The rated capacity (see 6.14.1 and 6.14.9 of MIL-PRF-29595) shall be not less than 50 ampere-hours at the end of life. To meet the capacity requirement, each battery shall produce a potential of not less than 20.0 volts at not less than 60 minutes when connected to a 50-ampere load after being tested in accordance with 4.5.8 as modified herein at 21°C (70 °F) (50 Ah/1.0 hr/21 °C/20.0 V at end of life). The battery's 1C-rate shall be 50 amperes."

3.9.12 Cycling. Delete the entire text and insert: "Batteries shall successfully complete not less than 1000 cycles for qualification inspection and 500 cycles for conformance inspection after being tested in accordance with 4.5.11 as modified herein. The battery shall produce a potential of not less than 20.0 volts at each 20-second duty cycle discharge. The battery shall also meet requirement (1) of table III of MIL-PRF-29595 when discharged in accordance with 4.5.11.f of MIL-PRF-29595. The inability of the battery to produce a potential of not less than 20.0 volts at each 20-second duty cycle discharge or meet requirement (1) of table III of MIL-PRF-29595 shall constitute a failure."

3.9.22 Output performance. Delete the entire text and insert: "The battery shall meet the minimum voltage requirements of figure 2 of this specification sheet for each profile discharge when tested in accordance with 4.5.21 of this specification sheet."

3.9.23 Discharge and charge at 71°C . Delete the text in the first sentence and insert the following: "The battery shall produce not less than 50 ampere-hours when discharged in accordance with 4.5.22.c of MIL-PRF-29595."

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3.9.24 Cold and hot temperature charge. Delete the first sentence and insert: "The battery shall produce not less than 30.0 ampere-hours after being tested in accordance with 4.5.23.1.c as modified herein. The battery shall produce not less than 37.5 ampere-hours in not less than 45 minutes after being tested in accordance with 4.5.23.1.h and 4.5.23.2.e as modified herein."

7.3 Add the following paragraph:

"3.9.28 Handles. The handles shall show no evidence of breaking away from the case or being damaged when tested in accordance with 4.5.40. The method of mounting handles, latches, and strikes shall be optional."

7.4 Modify the following subparagraphs:

4.2 Qualification inspection. Delete the text in the first sentence and insert the following: "Qualification inspection shall consist of the examinations and tests specified in table I of this specification sheet (see 6.4 as modified herein and 6.5 of MIL-PRF-29595)."

4.3.5 Group B inspection. Delete the text and insert the following: "The group B test shall consist of the examinations and tests specified in table I of this specification sheet (see 6.8 of MIL-PRF-29595)."

4.5 Methods of inspection. Delete the text in the first sentence and insert the following: "The tests on each battery shall be performed in the order as specified in table I of this specification sheet."

4.5.8 Capacity discharge. Delete the entire text and insert: "Stabilize the battery at room temperature (see 4.4.1 of MIL-PRF-29595). Discharge the battery at 50-ampere to a cutoff voltage of 20 volts. Constant potential charge the battery at 28 ±0.25 volts for 2 hours. Examine the battery for the requirements of 3.9.9 as modified herein and requirement (1) of table III of MIL-PRF-29595."

4.5.9 High low voltage charge test. In 4.5.9b, delete "22.0 volts" and insert "24.0 volts."

4.5.11 Cycling test. Delete the entire text and insert: "For qualification inspection, subject the battery to 1000 cycles of discharge-charge-open circuit and to the temperature environments in accordance with 4.5.11.b as specified herein. For conformance inspection, subject the battery to 500 cycles of discharge-charge-open circuit in accordance with 4.5.11.b as specified herein.

a. Preparation: Prepare the battery as follows:

(1) Constant potential charge the battery in accordance with 4.4.2.

(2) Rest the battery on open circuit for 1 ±0.1 hours.

b. Duty cycle: Cycle the battery as follows:

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(1) Discharge at a constant current rate of 1000 amperes for 20 seconds.

(2) Rest the battery on open circuit for 260 seconds.

(3) Discharge at a constant current rate of 1000 amperes for 20 seconds.

(4) Within 4 seconds after completing step 4.5.11.b(3), constant potential charge the battery at 28.25 ±0.25 volts for 1 hour 55 minutes.

(5) Rest the battery on open circuit for 1 ±0.1 hours.

c. Repeat step 4.5.11.b continuously for a total of 1000 duty cycles for qualification inspection and 500 duty cycles for conformance inspection.

d. Discharge at room temperature at the C-rate (50A) to a cutoff voltage of 20.0 volts.

e. Examine the battery for the requirements of 3.9.12."

4.5.18 Physical integrity at high temperature test. Delete the requirement of lifting the battery by the vent tubes of 4.5.18.e.

4.5.21 Output performance test. Delete and substitute: "The battery shall be subjected to the following test in the sequence shown. The battery shall meet the requirements of 3.9.22 for every profile discharge conducted.

"a. Charge the battery in accordance with 4.4.2.

"b. Place the battery in a -18 °C (0 °F) temperature chamber for 24 hours. The battery shall remain in the -18 °C (0 °F) environment for the duration of steps c, d, and e.

"c. Heaters may be activated. If heaters are used they shall be self-activated when the discharge is initiated. Discharge the battery at 50 amperes for 15 minutes. At the conclusion of the 15-minute discharge initiate the profile discharge of figure 2 of this specification sheet.

"d. After the profile discharge, rest the battery on open circuit for 4 minutes and then repeat the profile discharge of figure 2 of this specification sheet one time.

"e. Repeat step d one time.

"f. Examine the battery for the requirements of 3.9.22."

4.5.22 Discharge and charge at 71°C. Delete 4.5.22.c and insert the following:

"c. With the battery still in the temperature chamber at 71°C (160 °F), discharge the battery at the 1.0 C-rate to a cutoff voltage of 20 volts."

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4.5.23 Cold and hot temperature charge test. Delete the entire text and insert the following:

"4.5.23.1 Half-hour charge test. This test shall consist of the following steps:

"a. Constant potential charge the battery in accordance with 4.4.2.

"b. Open circuit the battery. Stabilize the battery (see 4.4.1) at -18 °C (0 °F).

"c. Keep the battery at the stabilized temperature of 4.5.23.1.b while discharging at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"d. Examine the battery for the requirements of 3.9.24.

"e. Open circuit the battery. Stabilize the battery at 25 °C (77 °F) (see 4.4.1). Keep the battery at 25 °C (77 °F) while constant potential charging the battery in accordance with 4.4.2.

"f. Repeat 4.5.23.1.b. Keep the battery at the stabilized temperature of 4.5.23.1.b while discharging it at the 1.0 C-rate to remove all but 25.0 ampere-hours as determined by 4.5.23.1.c.

"g. Repeat 4.5.23.1.b. Constant potential charge the battery for 30 ± 1.5 minutes in accordance with 4.4.2.

"h. Immediately after completing 4.5.23.1.g, place the battery in a 25° C (77° F) environment. Within 5 minutes, discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"i. Repeat 4.5.23.1.a through 4.5.23.1.h above but change the temperature of 4.5.23.1.b from -18 °C to 55 °C (0 °F to 130 °F).

"j. Examine the battery for the requirements of 3.9.24.

"4.5.23.2 One-hour charge test. This test shall consist of the following steps:

"a. Constant potential charge the battery in accordance with 4.4.2.

"b. Open circuit the battery. Stabilize the battery (see 4.4.1) at -40 °C (-40 °F). Perform 4.5.23.2.c and 4.5.23.2.d with the battery in the environmental chamber at -40 °C (-40 °F).

"c. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"d. Repeat 4.5.23.2.b. Constant potential charge the battery in accordance with 4.4.2 for 1.0 ±0.05 hour.

"e. Immediately after completing 4.5.23.2.d, place the battery in a 25 °C (77 °F)

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environment. Within 5 minutes, discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"f. Examine the battery for the requirements of 3.9.24."

4.5.26 Vibration test. Delete 4.5.26.d and insert the following: "Vibrate along each orthogonal axis (in the axis order shown on figure 3 of MIL-PRF-29595) in accordance with the test levels specified on figure 3 of this specification sheet. The battery shall be vibrated for a period of 2 hours along each axis."

4.5.38 Penetration test. Delete the text in the first sentence and insert the following: "The battery shall be tested in accordance with step (b)." Delete 4.5.38.b(3) and insert the following: "Samples shall be at a 100% state of charge before the test. Fire three 7.62mm API rounds through the battery in the direction specified above. At a minimum, one of the rounds shall be a tracer round."

4.5.40 Special tests. Delete the text and insert the following:

7.5 Add the following paragraphs:

"4.5.40.1 Handle test. Each handle, when swung out from the case against its stop, shall have a force of 600 Newtons (135 pounds) applied at the handle grasp and in a direction of maximum torque acting on the handle stop. The battery shall meet the requirements of 3.9.28."

4.5.41 Final examination. Delete the second sentence.

7.6 Modify the following subparagraphs:

6.4 Qualification. Insert the following after the first sentence: "Batteries must have successfully completed the safety test requirements of S9310-AQ-SAF-010 and received safety certification approval before placement on the QPL."

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TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examination and tests	Qualification Inspection							Conformance inspection		
		Sample number					Requirement paragraph 1/	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3	4	5				1	2
1	Incoming inspection	X	X	X	X	X	3.9.1 (G)	4.5.1 (G)	X	X	X
2	Visual and mechanical examination	X	X	X	X	X	3.9.2 (G)	4.5.2 (G)	X	X	X
3	Dimensions and weight	X	X	X	X	X	3.9.3 (G)	4.5.3 (G)	X	X	
4	Handles	X	X	X	X	X	3.9.28 (N)	4.5.40 (N)	X	X	
5	Electrical test preparation	X	X	X	X	X	---	4.5.7 (G)		X	X
6	Capacity discharge	X	X	X	X	X	3.9.9 (S)	4.5.8 (S)	X	X	X
7	High and low voltage charge	X					3.8.1 (G) 3.9.10 (G)	4.5.9 (G)			
8	Charge and discharge test at low temperature				X		3.9.9 (G) reqmt (3) table III (G)	4.5.13 (G)			
9	Discharge while inverted		X				3.6.2 (G), reqmt (4) table III (G)	4.5.14 (G)			X
10	Cycling	X					3.9.12 (S)	4.5.11 (S)		X	
11	Half-hour charge		X				3.9.24 (S)	4.5.23.1 (S)			
12	One-hour charge			X			3.9.24 (S)	4.5.23.2 (S)			
13	Altitude		X				3.8.1 (G) 3.8.13 (G)	4.5.30 (G)			X
14	Electromagnetic interference				X		3.9.13 (G)	4.5.12 (G)			
15	Deep discharge recovery			X			3.9.20 (G)	4.5.20 (G)			
16	Shock test (basic design)			X			3.8.1 (G) 3.9.13 (G)	4.5.28 (G)		X	
17	Output performance test		X				3.9.22 (S)	4.5.21 (S)			X
18	Temperature shock test	X					3.8.1 (G), 3.9.13 (G)	4.5.33 (G)		X	
19	Vibration test			X			3.8.1 (G), 3.9.3 (G), 3.9.25 (G)	4.5.26 (S)		X	
20	Sand and dust test					X	3.8.1 (G) 3.9.8 (G) 3.9.9 (G)	4.5.35 (G)			
21	Humidity test		X				3.8.1 (G), 3.9.13 (G)	4.5.31 (G)			X
22	Salt fog test	X					3.8.1 (G), 3.9.13 (G)	4.5.32 (G)		X	
23	Temperature rise and float test		X				3.9.15 (G)	4.5.15 (G)			X

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TABLE I. Qualification inspection and conformance inspection of batteries - Continued.

Test number	Examination and tests	Qualification Inspection							Conformance inspection		
		Sample number					Requirement paragraph 1/	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3	4	5				1	2
24	Ground storage					X	3.9.18 (G)	4.5.36 (G)			
25	Physical integrity at high temperature test				X		3.8.1 (G), reqmt (1) table III (G)	4.5.18 (S)			
26	Shock test (crash safety)			X			3.9.13 (G), 3.9.27 (G)	4.5.29 (G)			
27	Discharge and charge at 71°C				X		3.9.23 (S)	4.5.22 (S)			
28	Shelf life					X	3.9.12 (G), 3.9.19 (G)	4.5.37 (G)			
29	Impact resistance test				2/		3.9.21 (G)	4.5.19 (G)			
30	Penetration test		X				3.9.16 (G)	4.5.38 (S)			
31	Crush test			X			3.9.17 (G)	4.5.39 (G)			
32	Final examination	X	X	X	X	X	3.9.4 (G)	4.5.41 (G)		X	X

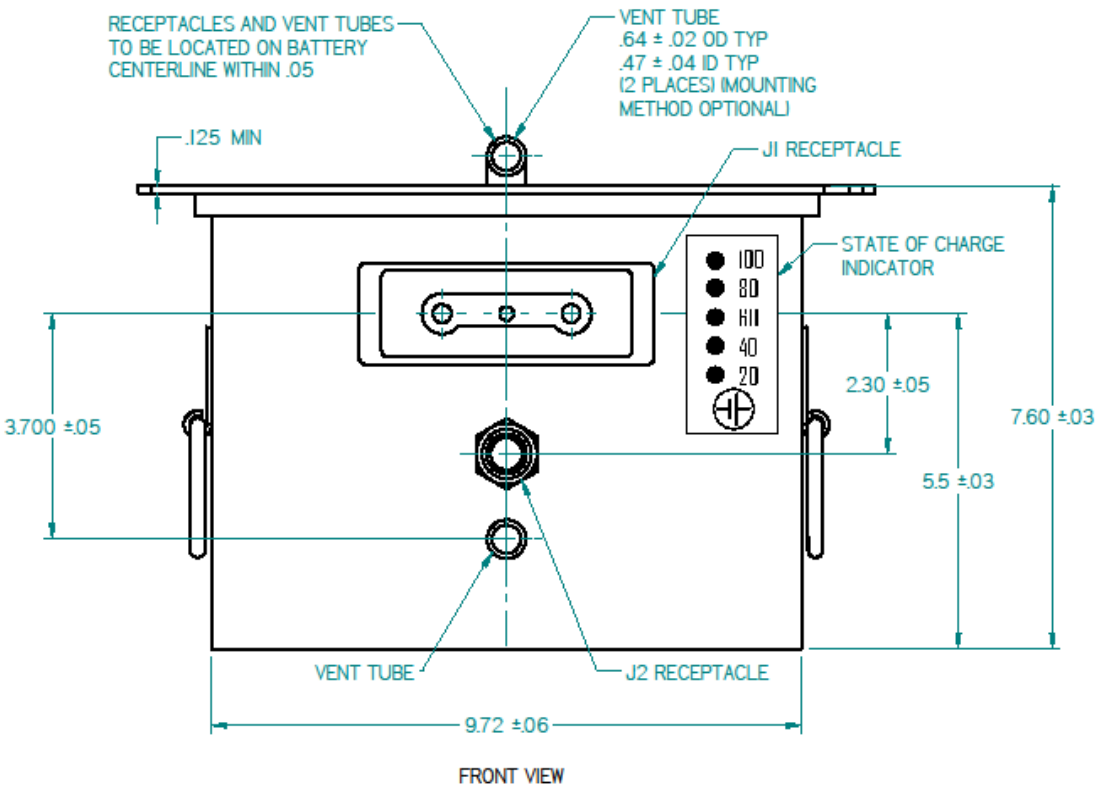
1/ (G) indicates that paragraph is in MIL-PRF-29595. (S) Indicates that paragraph has been modified by specification sheet. (N) Indicates new requirement added by specification sheet.
2/ For nonmetallic battery container designs, two nonmetallic battery containers shall be furnished for inspection.

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TABLE II. Safety test inspection (S9310-AQ-SAF-010 requirements) list.

Test number	Examination and tests	Sample number	Requirement paragraph	Test paragraph
1	Penetration test	1 each minimum	3.9.16	4.5.38
2	Crush test	1 each minimum	3.9.17	4.5.39
3	Short Circuit Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
4	Overcharge/Discharge Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
5	Overdischarge/Charge Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
6	High Temperature Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
7	Electrical Safety Device Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
8	Aging Safety Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010

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NOTES:
1. All dimensions are in inches.
2. Abbreviations used in this figure are inside diameter (ID), maximum (MAX), minimum (MIN), outside diameter (OD), part identification number (PIN), requirement (REQ), and typical (TYP).

FIGURE 1. Dimensions and configuration for PIN M29595/2-1.

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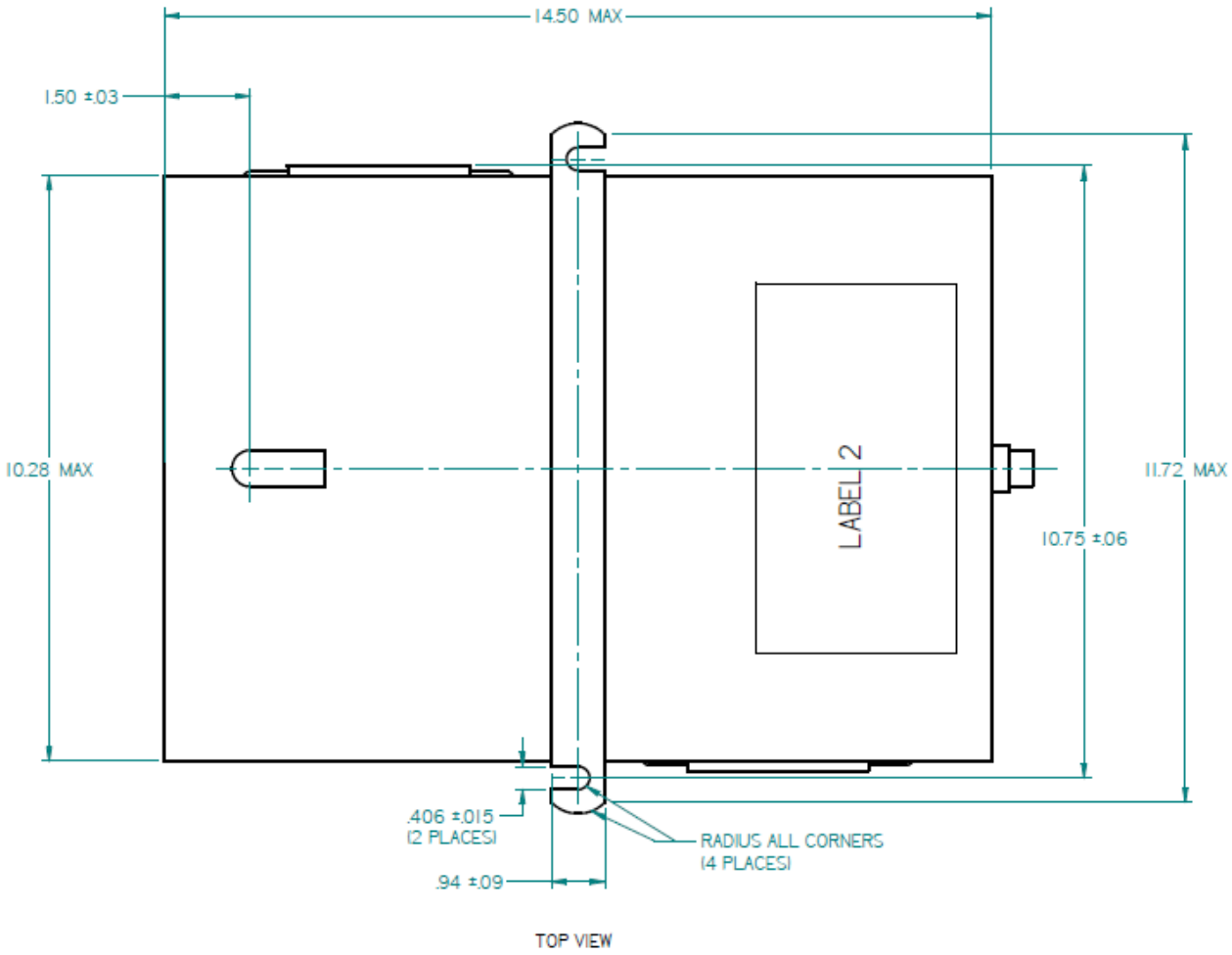


FIGURE 1. Dimensions and configuration for PIN M29595/2-1 – Continued.

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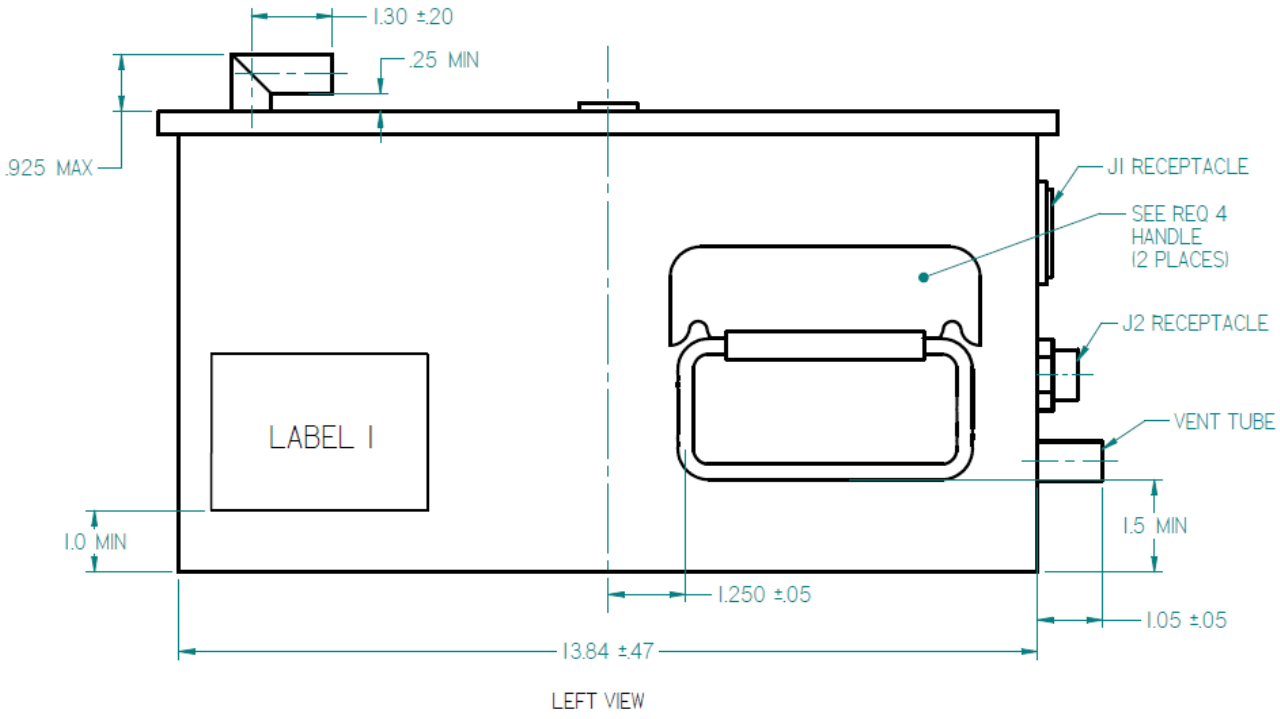


FIGURE 1. Dimensions and configuration for PIN M29595/2-1 - Continued.

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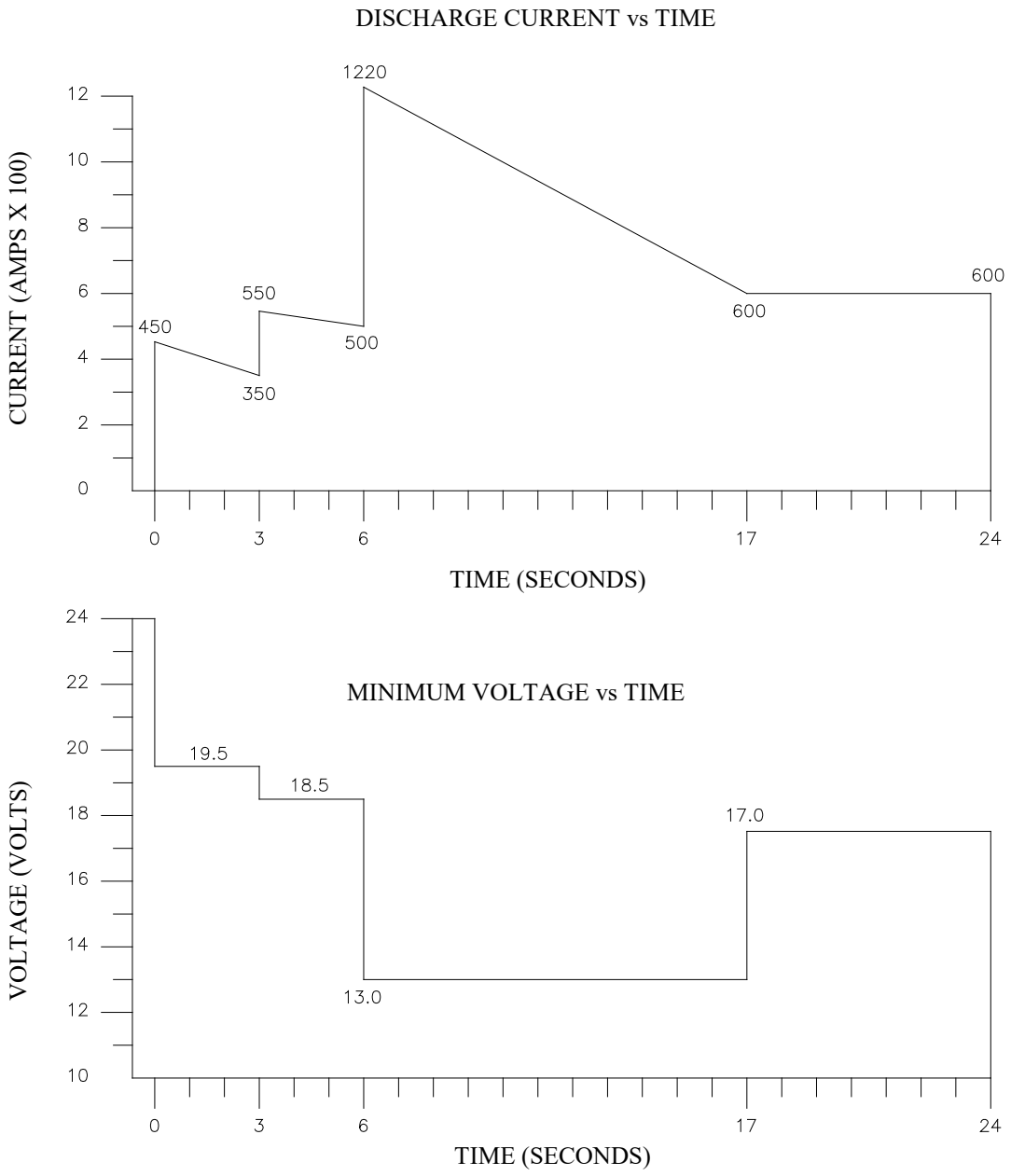


FIGURE 2. Profile discharge at -18 °C (0 °F).

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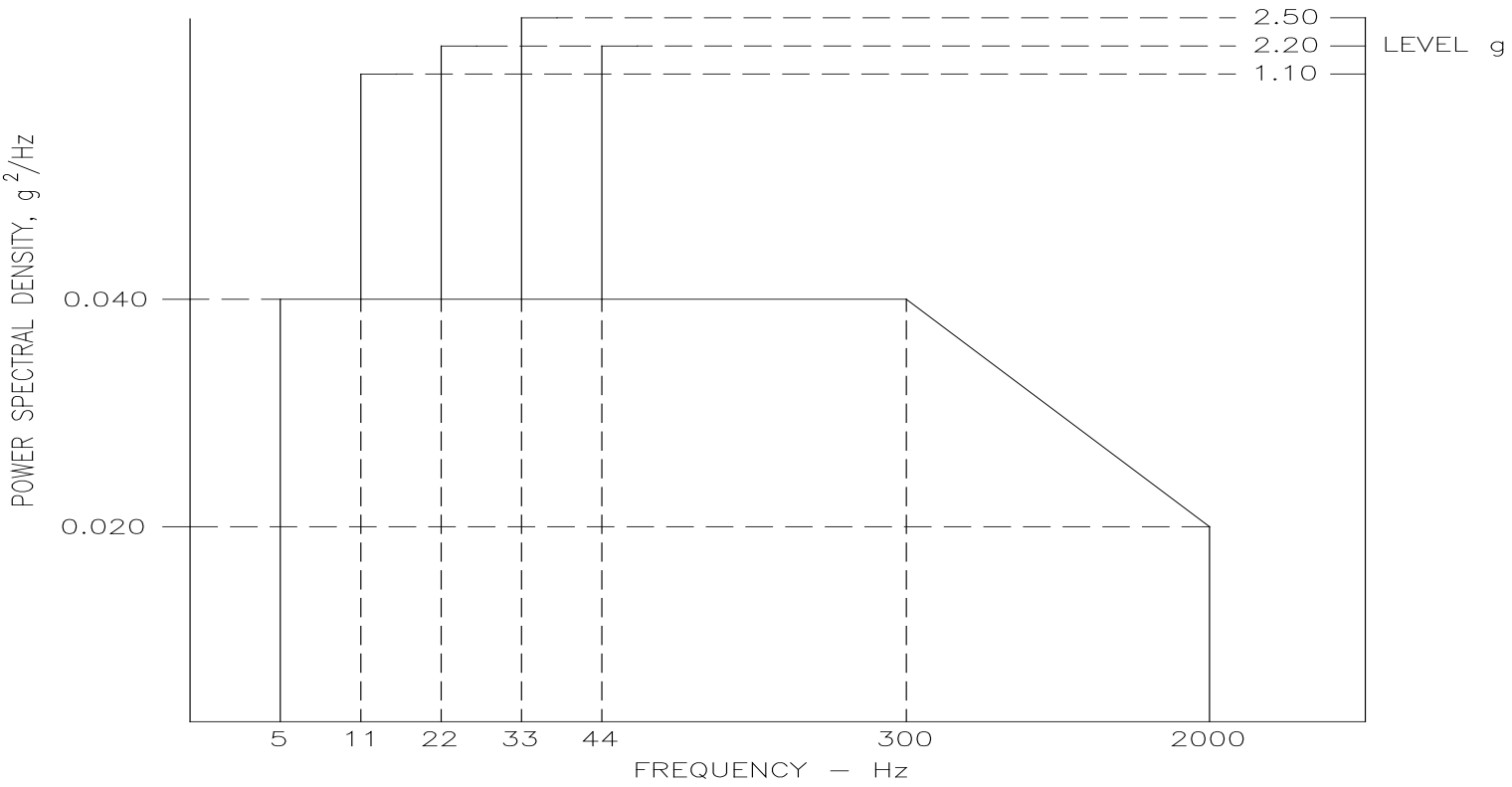


FIGURE 3. Vibration profile.

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Referenced documents. In addition to MIL-PRF-29595(AS), this specification sheet references the following documents:

MIL-DTL-26482
MIL-PRF-18148/3
SAE AMS-STD-595 color 17875
S9310-AQ-SAF-010

Custodian:
Navy – AS

Preparing activity:
Navy – AS
Project 6140-2023-001

Agent:
Navy – SH

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notation.

NOTE: The activities listing above were interested in this document as of the date of the document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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